

frequency count method

Algo Sum(A, n)

```

1  S = 0
2  for (i=0; i<N; i++)
3      S = S + A[i]
4  return S

```

$n = 5$

Diagram illustrating the frequency count method for the loop `for (i=0; i<N; i++)`. The loop runs for n iterations. The frequency of each iteration is counted as 1. The total frequency is $n+1$ (including the initial state). The total time complexity is $O(n)$.

Handwritten calculation:

$$f(n) = 1 + n + 1 + n + 1 = 3n + 3 = O(n)$$
space complexity

variable used

A — n

n — 1

S — 1

i — 1

$$S(n) = n + 1 + 1 + 1$$

$$= n + 3$$

$$O(n)$$

Example 2

sum of two matrix

sum(A, B, n)

2

for (i=0; i<n; i++) — n+1

{

for (j=0; j<n; j++) — n

{

C[i,j] = A[i,j] + B[i,j]

}

}

3

3

space

variables

A — n^2 B — n^2 C — n^2

n — 1

i — 1

j — 1

$$S(n) = 3n^2 + 3$$

$$= O(n^2)$$

ex multiplication of 2 matrix

multiply(A, B, n)

2

for (i=0; i<n; i++) — n+1

{

for (j=0; j<n; j++) — n(n+1)

{

C[i,j] = 0 — n — n

for (k=0; k<n; k++) — n — n — (n+1)

{

}

}

}

$$C[i,j] = C[i,j] + A[i,k] \times B[k,j] \quad - n \dots$$

3

$$\begin{aligned} S(n) &= (n+1) + n(n+1) + n^2 + n \times n \times (n+1) + n^3 \\ &= n+1 + n^2 + n + n^2 + n^3 + n^2 + n^3 \\ &= 2n^3 + 3n^2 + 2n + 1 \\ &= O(n^3) \end{aligned}$$

space

A $\rightarrow n^2$
B $\rightarrow n^2$
C $\rightarrow n^2$
n $\rightarrow 1$
i $\rightarrow 1$
j $\rightarrow 1$
k $\rightarrow 1$

$O(n^2)$

other Episode - 6

for (i=1; i<n; i+=2)

{ statement $\rightarrow \frac{n}{2}$

}

$$S(n) = \frac{n}{2}$$

$O(n)$

① for (i=0; i<n; i++) $\rightarrow n+1$
{
 for (i=0; i<n; i++) $\rightarrow n$
 { statement; $\rightarrow n$
 }
}

$$\begin{aligned} &= 1+2+3+\dots+n \\ &= \frac{n(n+1)}{2} \\ &= \frac{n^2+n}{2} = O(n^2) \end{aligned}$$

i	j	no. of times
0	0x	0
1	0x 1x	1
2	0x 1x 2x	2
3	0x 1x 2x 3x	3
...	...	...
n	...	n

② P=0
for (i=1; P<n; i++)
{
 P=P+i

}

assume P > n its stopping

$$\text{Since } P = \frac{k(k+1)}{2}$$

$$\begin{aligned} 1 &= 1 \\ 2 &= 1+2=3 \\ 3 &= 1+2+3 \\ 4 &= 1+2+3+4 \\ &\vdots \\ &= 1+2+3+\dots \end{aligned}$$

$$\therefore \frac{k(k+1)}{2} > n$$

$$= k^2 > n$$

$$= k > \sqrt{n}$$

$$O(\sqrt{n})$$

Episode - 7

Q for (i=0; i<n; i=i*2)
{ statement
}

1

$$1 \times 2 = 2$$

$$2 \times 2 = 2^2$$

$$2^2 \times 2 = 2^3$$

1

$$2^k \times 2 = 2^{k+1}$$

Assume $i \geq n$ then it stops

$$\text{since } i = 2^k$$

$$2^k \geq n$$

$$k (\log_2 2) \geq \log_2 n$$

$$k \geq \log_2 n$$

$$O(\log_2 n)$$

Another way to Analyse

for (i=0; i<n; i++)

{ statement

}

$$i = 1 + 1 + 1 + 1 + \dots = n \rightarrow \text{steps at } n$$

$$O(k) = O(n)$$

for (i=0; i<n; i=i*2)

{ statement

}

$$i = 1 \times 2 \times 2 \times 2 \dots = n \rightarrow \text{steps at } n$$

$$2^k = n$$

$$k \log_2 2 = \log_2 n$$

$$O(\log_2 n)$$

Now say this if $n=8$ and $n=10$

$$\log_2 8 \Rightarrow \log_2 2^3 \Rightarrow 3 \log_2 2 \Rightarrow 3 =$$

$$\log_2 10 \Rightarrow 3.2$$